

KEYWORDS

Keywords (max. 5)

soil heath and fertility, potato quality, potato pathogens, biological control of plant diseases, cropping systems research.

PROJECT SUMMARY

The European Union is committed to transforming its agri-food systems to be more environmentally sustainable, climate-friendly, and resilient. A key strategy for achieving these goals is the use of crop diversification, specifically the incorporation of winter cover crops into potato rotations. However, the benefits of cover crops for potato cultivation are not fully recognized or exploited in some countries, particularly in the major production areas of northwest Europe.

This project aims to address this gap by determining effective cover crop strategies for potato production in northwestern countries (EE, SE, NO, NL and DE) and providing evidence-based communication for their use. To achieve this, field experiments will be conducted across different locations and climate conditions in research farms and through farmer-led field trials (living labs). These experiments will evaluate both established cover crops and novel plants suitable as cover crops, assessing their impact on soil functions (including pathogen load), disease incidence, the integration of microbial biocontrol agents, plant performance, soil microbial community diversity, and potato yield.

At the start of the project, workshops surveying both researchers and practitioners will be held to identify knowledge gaps, beliefs, and communication needs among the participants. The project will use this information in a co-creation, multi-actor approach, involving all relevant stakeholders in the design, implementation, and evaluation of the experiments to ensure the project's activities are relevant and feasible. We will also strive to ensure socio-economic factors are considered for end users. In this way we will also maximise the dissemination of the results and create a positive impact on the agricultural community. Our transnational approach brings together stakeholders and researchers from EE, & SE (where a few early adopters use cover crops) and NO (where cover crops are not yet used) with more experienced cover crop users in NL and DE. Our focus stakeholders in SE are starch potato producers, whilst in NO we will work closely with table potato producers. Combining knowledge from these groups and the current users in NL and DE will allow us to broaden perspectives and impacts and will drive innovation within the project.

To showcase the benefits of cover crops and encourage their adoption, demonstration plots that will serve as lighthouses, and field days will be organized, along with a discussion and exchange symposium for farmers, advisors and other interested stakeholders. Based on the project findings, recommendations will be developed for the practical use of cover crops in potato rotations, highlighting their adaptability to different potato production systems in northwestern Europe. Our interdisciplinary team combines both natural sciences and humanities approaches within agricultural sciences, including soil chemistry, molecular plant-microbe interactions, plant sciences, soil microbiology, agronomy and communication research along with advisory services and practical farming knowledge to achieve these outcomes.

By promoting the use of cover crops, this project will contribute to the EU's goals of sustainable and climate-friendly agriculture. The findings will provide valuable data for both national and EU level agri-food policies, and the increased awareness and adoption of cover crops will benefit the agricultural community, consumers, and the environment.

DESCRIPTION OF WORK

EXCELLENCE

1.1 Objectives

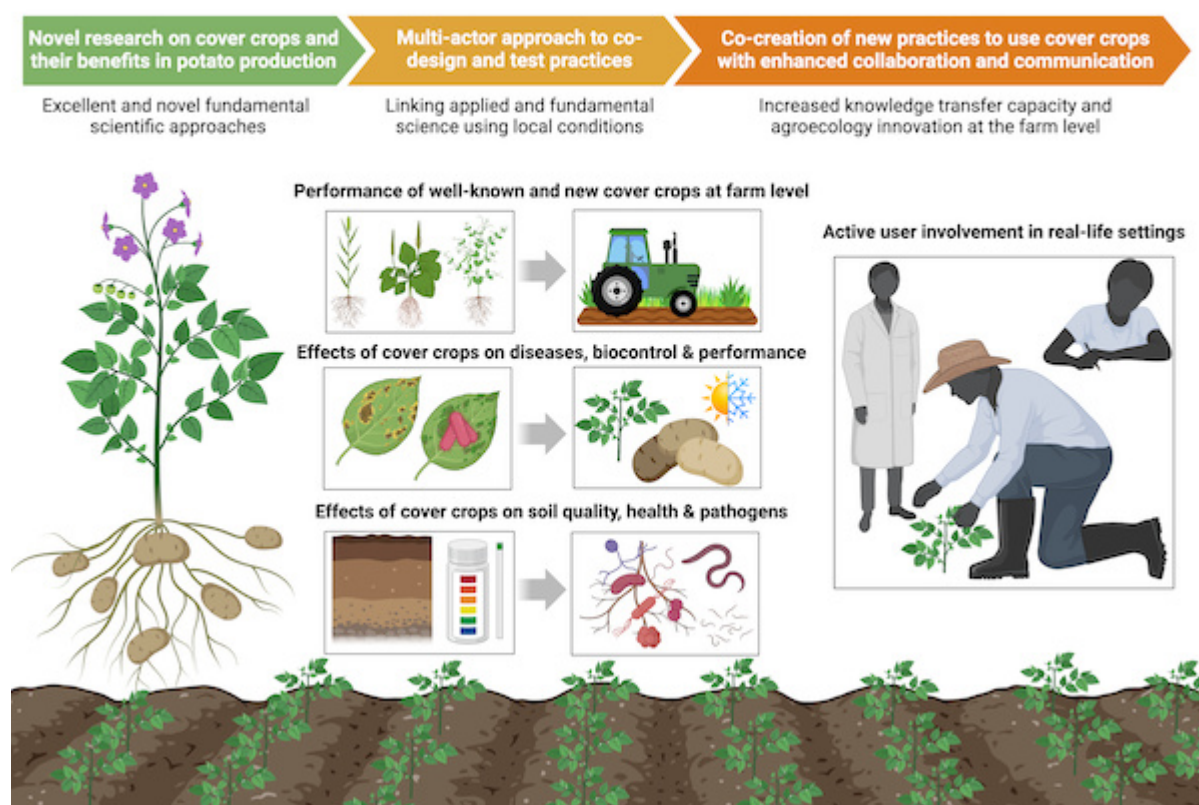
DisCoverCrop aims to facilitate crop diversification for sustainable potato farming in Europe with the successful adoption of cover crops (Fig.1). Potato production faces several current and future challenges such as, increasing disease and pest pressure, stress related to climate change, as well as leaching of nutrients along with soil compaction and erosion in intensive production systems. Therefore, there is an urgent need for tailored sustainable agronomic practices along with region-specific and robust practical guidelines for European potato cropping systems. By diversifying potato crop rotation with well-established, but also, novel cover crops that consider the local conditions in the major potato-growing regions of Europe, DisCoverCrop will strengthen resilience and sustainability in the potato production value chain

and enhance overall productivity and economic viability.

Our objectives are:

1. To assess the role of currently used and novel cover crops in improving soil health whilst also reducing erosion, maintaining carbon storage, optimizing nutrient cycles and reducing the use of contentious chemicals in European potato production.
2. To assess how cover crops can promote the natural regulation of potato diseases and improve both the performance and integration of microbial biocontrol agents within potato cropping practices.
3. To evaluate the social considerations, local knowledge, concerns, and decision-making strategies of potato farmers working with cover crops, to support the development of new agroecological management practices and economic gains for the use of cover crops at the farm level.
4. To facilitate knowledge exchange between scientists, farmers and other stakeholders regarding common potato production practices, issues, and solutions, to enable the implementation of cover crops, integrated with biocontrol methods where possible, in practice.

DisCoverCrop



1.2 State of the Art

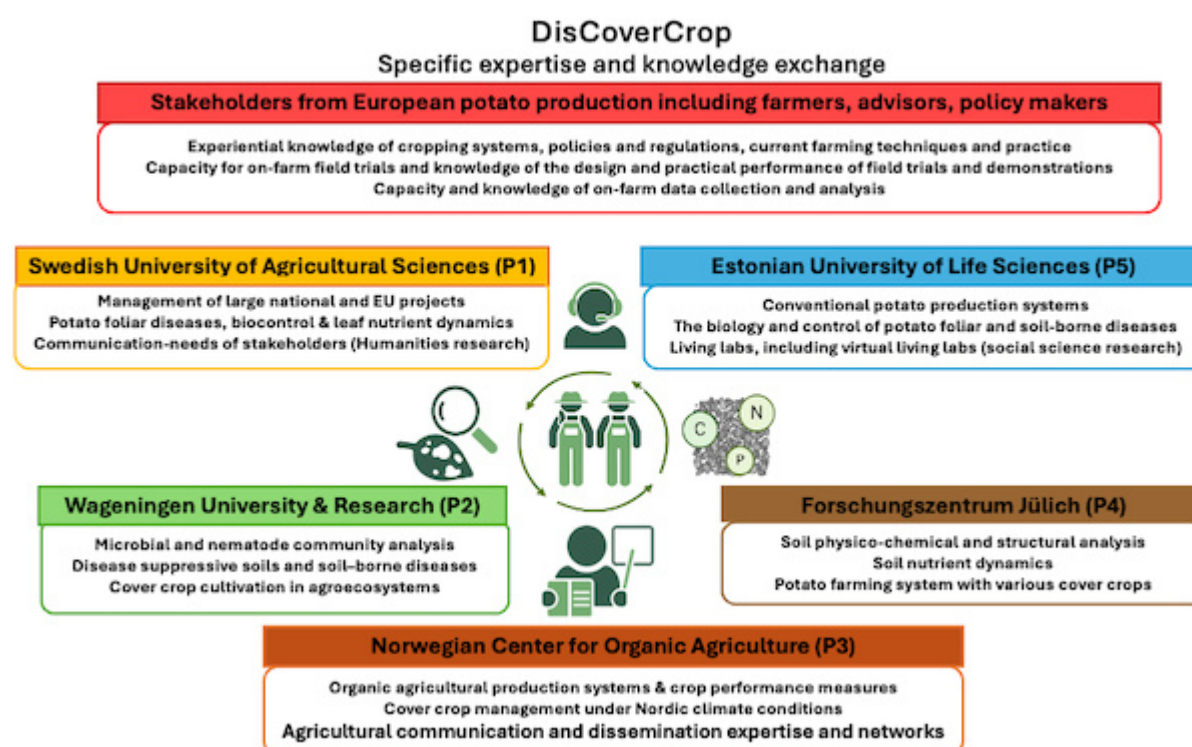
Potato (*Solanum tuberosum*) is one of the top 5 food crops worldwide, (global production >350Mt in 2020) (FAO, 2020). In Europe, significant contributions come from DE, FR, and NL, collectively responsible for 40% of the global harvest, with the highest yields worldwide of up to 42 tha⁻¹ (Devaux et al., 2020; Eurostat, 2022). However, as the focus is on short-term yields, sustainable practices such as cover cropping are often neglected, leading to a deterioration of soil health in these areas (Goffart et al., 2022).

Potato production systems heavily rely on mineral fertilizers, pesticides, intensive tillage, and short crop rotations. Depending on the country, potatoes are grown in a multi-year rotation (DE, NL) or in more narrow rotations (EE, SE), whereas in NO they are typically grown year after year on the same land. Over time, this leads to soil fertility decline and compromises long-term productivity (Goffart et al., 2022; Powell et al., 2020). This leads to reduced soil organic matter, diminished water holding capacity, compromised soil structure, and decreased biodiversity and fertility (Cardoso et al., 2013). Consequently, potato diseases may lead to potential risks to food security (Abawi and Widmer, 2000).

Cover cropping emerges as a promising solution, widely adopted in agri-food systems, to address these challenges (Larkin et al., 2021; Cazzaniga et al., 2023). The use of cover crops during fallow periods or alongside catch crops provides a continuous vegetative cover that helps to prevent nutrient and organic carbon losses, counteracts soil erosion and compaction and increase soil biodiversity EEA (2022). Integrating cover crops into potato rotations enhances soil physio-chemical characteristics, mitigates disease pressure, enriches microbial communities, and improves long-term yields and harvest quality (Collins et al., 2006; Larkin et al., 2010; Hemkemeyer et al., 2024; Rittl et al., 2023). However, despite their recognized benefits, widespread adoption across Europe faces challenges. Northern countries like EE, SE and NO are hindered by short growing seasons, while although Western countries like DE and NL, have established practices, they lack practical guidance for optimal integration. Addressing these barriers is crucial to unlock the full potential of cover crops to enhance the sustainability and productivity of European potato production.

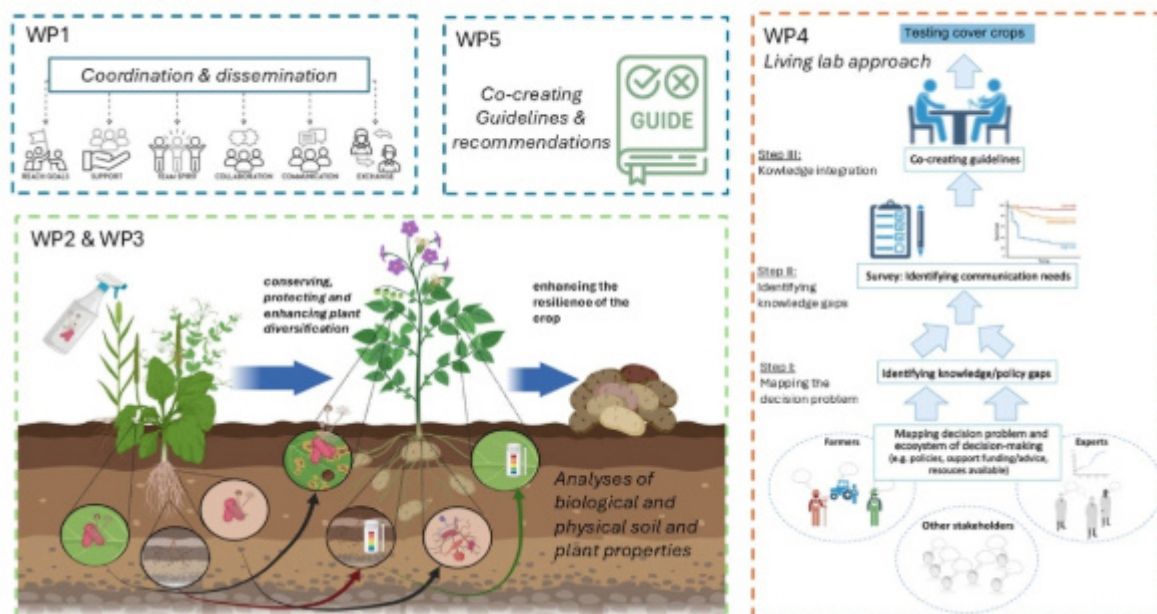
1.3 Concept and Approach

DisCoverCrop comprises a multidisciplinary team of partners with expertise in the areas of soil science, plant science, soil microbiology, integrated plant protection, communication research, living labs and agricultural advisory strategies in both, conventional and organic agriculture that integrates natural, humanities and social sciences (Fig. 2). This project consortium will enhance knowledge transfer from scientific evidence into the field, to the farmer to understand both their beliefs and potential misconceptions and solve some of the uncertainties potato farmers are confronted with e.g., how to utilize cover crops on their farms. It will also allow farmers and advisors to transfer knowledge to the scientific community on the challenges they face in implementation. Bilateral knowledge exchange from practitioners to researchers will allow us to design research with practical outcomes in real-world settings (Fig.3).



Our project is divided into 5 workpackages focusing on farm-managed field trials in SE, NO, DE, EE and NL that are co-designed in a multi-actor approach. We will assess performance, development, biomass and rooting properties of cover crops and relate the findings to the overall potato health and yield. Soil sampling in the spring and at potato harvest will allow us to investigate induced changes in physio-chemical soil properties and interactions with common pathogens, to uncover the underlying mechanisms of cover cropping that contribute to improved soil and crop health. Abiotic and biotic characteristics (below and aboveground) will be compared, and research field trials will include experiments to test the ability of cover crops to host biocontrol agents and to increase microbial biodiversity.

DisCoverCrop – Main methodologies and outputs



1.4 Ambition

European potato farmers show a willingness to adopt sustainable practices, as evidenced by reports from the European Potato Processors Association (EUPPA, 2021) and commitments from companies like McCain (2024). However, a successful transition relies on accessible, evidence-based guidelines. Discovering differences in requirements for the use of cover crops in varying climatic regions in NW Europe (Fig. 4) will be a key aspect of our research and the basis of our recommendations for practitioners. Our ambition is to work with farmers to refine cover crop strategies that work with farming practices to increase their use at the farm-level to promote sustainability in (potato) production. The knowledge gained will help shape more sustainable farming practices tailored to NW European agroecosystems. By promoting the integration of cover crops in EE, SE and NO and providing evidence of the benefits derived from using (novel) cover crops in DE and NL, we aim to catalyse the adoption of resilient potato cropping systems to local challenges across Europe.

1.5 Added value for transnational research and innovations

DisCoverCrop aims to develop evidence-based recommendations for the adoption of sustainable cover crop practices in farming. An interdisciplinary framework and discussion platform will be established to facilitate international collaboration and knowledge exchange among potato farmers and experts. Our research is connected to ongoing projects working with cover crops on the European scale e.g., "Terra Nordica", "CropDrive", "CoverCropNordica" and national projects on cover crops, biocontrol and soil health. Farmers and researchers will benefit from sharing practical and scientific knowledge between the different countries that face both similar and different production challenges.

DisCoverCrop – Partner country overview



RELEVANCE

2.1 Relevance to the call scope and main objectives

Cover crops are important aspects of sustainable agroecological cropping practices, however, more research is needed to facilitate their uptake particularly in Northern Europe.

DisCoverCrop directly addresses the call's thematic area of agroecological innovations on the farm-scale. It focuses on the subtopics of improving soil health with cover crops (1.1), promoting plant diversification and rotations with cover crops in potato cropping systems (1.2), developing new agronomic practices to increase potato cropping systems' resilience, efficiency, and sustainability (1.3) and promoting natural regulation of pests and diseases, along with the integration of biocontrol practices within potato cropping systems (1.5). Our multi-actor interdisciplinary approach that puts the practitioner at the centre of the research project is fully in line with the objectives of Agroecology Europe and this call, by bringing together farmers, researchers, and other relevant stakeholders to co-develop, test and monitor new practices, approaches and technologies for the use of cover crops in real-life contexts (Fig. 1, 2). Our on-farm field trial approach will establish a new network of living labs with active user involvement in real-life settings involving multi-method approaches, collaborations between multiple stakeholders, and co-creation through co-design, co-assessment and co-implementation of research and innovation. By working across different countries in Europe, we will adapt our solutions to local challenges, whilst at the same time benefiting from an exchange of experiences and ideas (Fig. 3, 6).

Our research will lead to actionable knowledge, operational solutions and agroecological innovations that are directly applicable at the farm level (Fig 5). Currently the agroecology toolkit largely overlooks the role and use of biocontrol in cropping systems, with many practitioners unaware of its value even though many of agroecology principles also support the survival and performance of biocontrol organisms. One of the innovative aspects of DisCoverCrop is that we will combine the use of cover crops with biocontrol to provide a holistic approach to the conservation of both soil and plant health in potato cropping systems.

2.2 Expected impacts

The outputs of the project will be new scientific and practical knowledge on the use and benefits of cover crops in potato production systems along with practical guidelines for their use that are tailored to local conditions. These outputs will lead to outcomes that include an increased understanding and use of cover crops in potato farming, a reduction in the use of synthetic inputs, improvements in soil health and biodiversity conservation, economic benefits, and increased knowledge exchange among stakeholders (Fig. 5).

The specific impacts on farming practices are likely to differ between the participating countries. In EE, SE and NO we expect that the project will contribute to the adoption of cover crop practices where now there are very few early adopters (EE, SE) or currently none (NO). This will be achieved by investigating (novel) cover crops that are suitable for Northern climates and by directly involving farmers in their implementation. In these countries, cover cropping will increase overall diversity and minimize soil erosion.

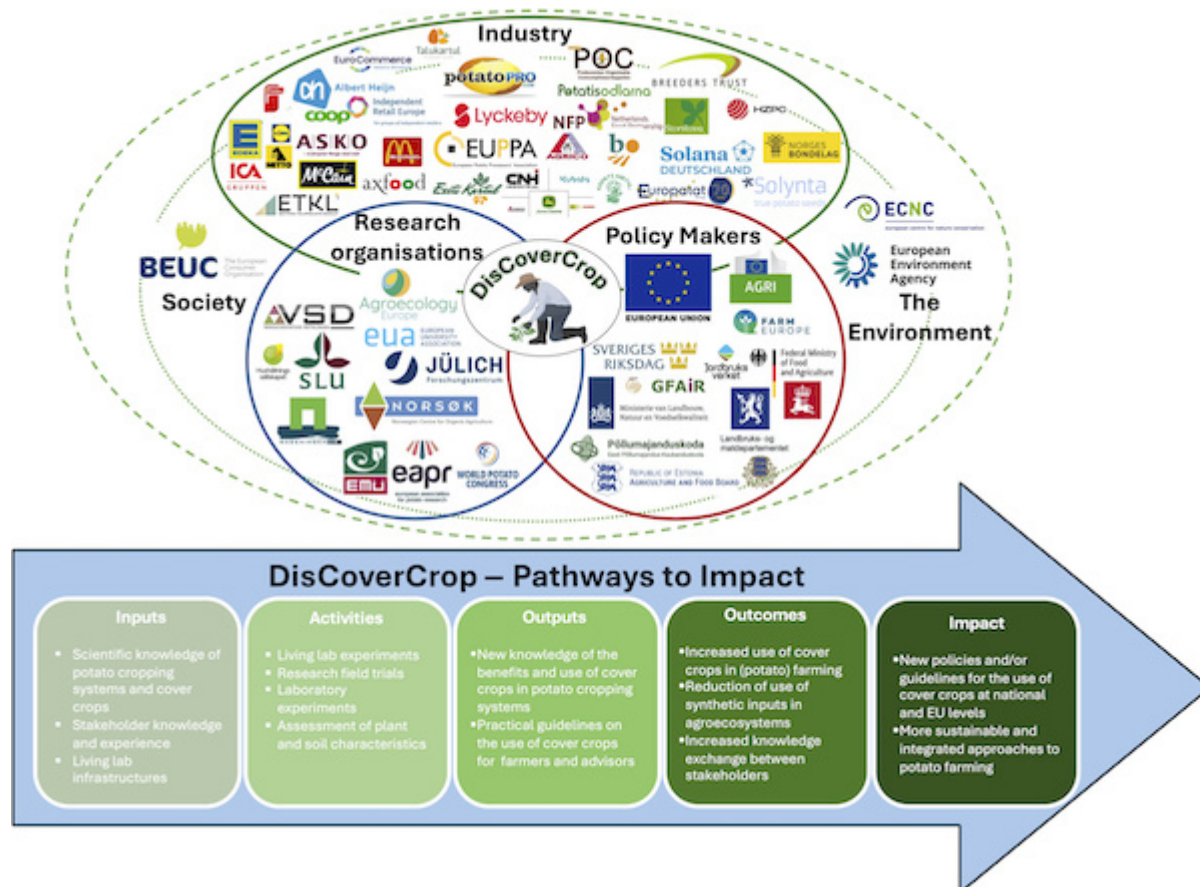
In NL and DE, where cover crops are already widely used, we expect to contribute to the more targeted use of suitable cover crops to meet the local challenges (i.e. nutrient leaching and pathogen pressure). This can result in reduced fertilizer and pesticide use and therefore reduced leaching into the environment.

Shorter-term research outputs such as scientific articles and other forms of scientific communication will impact the research field, driving innovation, new collaborations and new scientific research. Longer term impacts will contribute to more sustainable farming practices which will be valorised in terms of new policies and guidelines for the use of cover crops at national, regional and European levels.

2.3 Measures to maximise impact

Since research shows that stakeholder mapping can enhance the co-creation and impact of research involving multiple actors (Skarlatidou et al., 2019), we have mapped DisCoverCrop stakeholders, using a quintuplet helix approach (upper panel, Fig.5) considering i) academia (blue circle), ii) industry (solid green circle), iii) policy makers (red circle), iv) society (dotted green circle), v) the environment (dashed green circle). Identifying stakeholders early on ensures that their knowledge and experience will be involved from the beginning of the project. This guarantees that research activities are relevant to practice and that results are directly applicable on the farm-level.

Research activities will be carried out in living labs together with close involvement of farmers, using their expertise at every step of the project. Specific research questions and experiments will be co-created together with input from researchers from different scientific disciplines, farmers, and advisors based on knowledge gaps identified and agreed on by both scientists and practitioners. By fostering collaborative synergies, the project aims to achieve outcomes that exceed those that could be accomplished through isolated national efforts.



2.4 Dissemination and Communication activities and exploitation of results

DisCoverCrop spans multiple, interdisciplinary topics that will generate large, diverse data sets that need to be easily accessible by all project partners. The unique and cross-disciplinary approach will therefore pose specific demands on data management, including tight interdisciplinary guidance, training, and infrastructures for domains of heterogeneous data according to FAIR principles (Findable, Accessible, Interoperable, and Reusable). We will ensure that we adhere to national and EU ethics and data management guidelines including GDPR rules for the handling and storage of personal data collected from farmers.

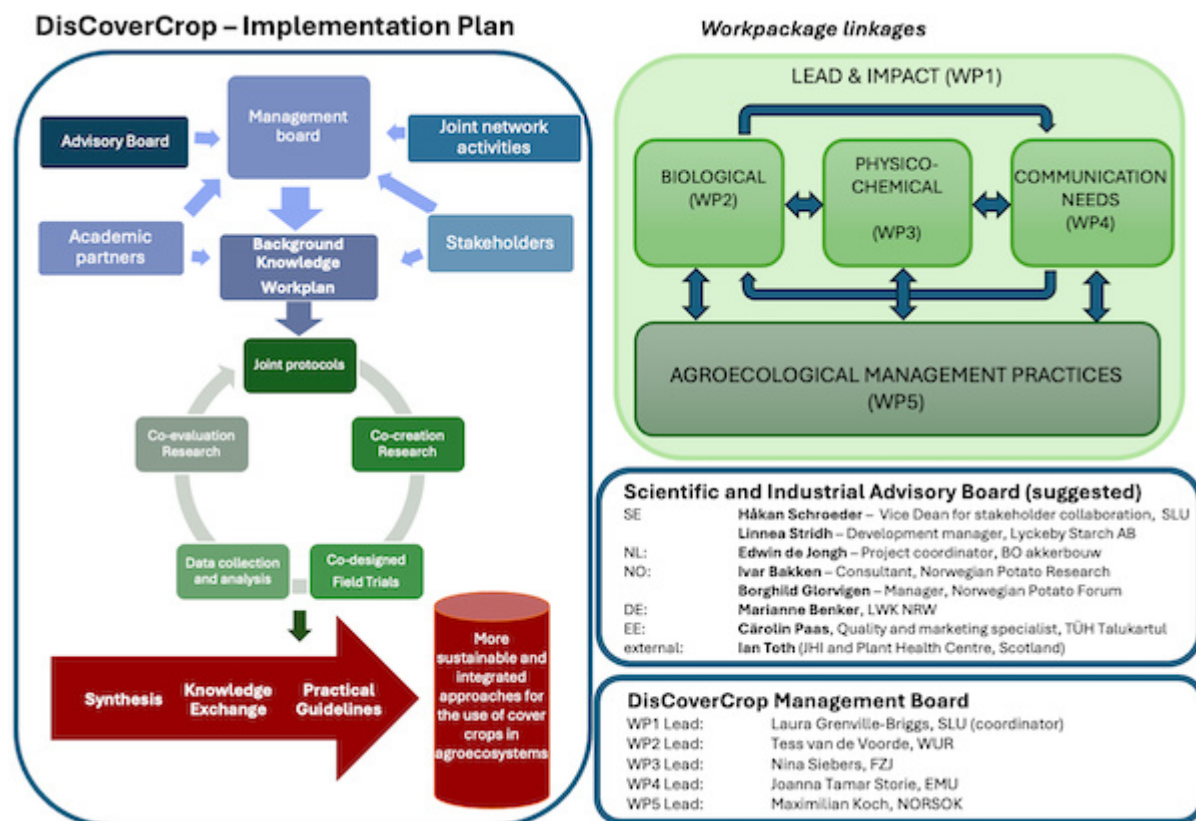
Effective communication involves more than simply relaying scientific information; it necessitates a dialogue that acknowledges the insights of local experts and farmers (Fischhoff et al., 2011; Persson et al., 2018; Labeyrie et al., 2021). Truly effective communication empowers recipients to make well-informed decisions (cf. Fischhoff et al., 2011). While the interaction between scientific and experiential evidence varies among cultures and professions, local experiential knowledge can be as influential as scientific evidence, shaping perspectives irrespective of regulatory frameworks (Ingram 2008; Blennow et al., 2020; Persson et al., 2022). Bridging the gap between decision-makers' existing knowledge and what they need to know is essential to enhance communication effectiveness (Fischhoff et al., 2011). We will therefore develop evidence-based guidelines for communication where communication needs evidence will be collected from our workshops and surveys. Our outreach channels include webinars, farm days, short videos, social media platforms and popular science articles.

We will ensure that published papers, reports, videos, and webinars are open access. Project updates will be shared on the project website every 4 months. Peer-reviewed papers or technical reports will be accompanied by a policy brief/fact sheet in English and the official languages of the partner countries. Digital platforms and a virtual living lab will be the primary means of communication with our stakeholders. The synthesis of relevant agronomic measures will be presented in a final online workshop that is open to all. The field experiments conducted by DisCoverCrop will serve as a platform for knowledge sharing and idea exchange among stakeholders, functioning as a living lab and small-scale research trials will be used as lighthouses to showcase best practice.

IMPLEMENTATION

3.1 Network organization and management structure

Our network comprises 5 academic institutions with complementary interdisciplinary expertise (Fig 2) along with associated stakeholders (see letters of interest and intent and description in the workplan). An overview of our implementation plan and workflow is shown in Fig 6a.



Our organisational structure is designed to facilitate knowledge-exchange and co-creation of new knowledge. Prof Laura Grenville-Briggs (LGB) at P1 (SLU) will coordinate the project establishing a consortium agreement and ensuring adequate progress and reporting. She has extensive experience coordinating large national and international grants, including EU projects. She will be supported by a management board (MB). The MB consists of the work package (WP) leaders, (which represent all 5 academic partners) and may also include task leaders as and when needed. WP leaders have been chosen in relation to their specific area of expertise and competence (Fig 2). In the MB decisions will be taken by a simple majority (50%+1). The implementation of the project will be governed by a consortium agreement, which will be drawn up, signed and come into force within the first month of the start of the project. Decision making within the general assembly including all researchers and stakeholders involved in the project which will meet online every 6 months will be taken by a majority of 2/3 of the votes cast. We will take input from an external advisory board as well as from joint network activities that LGB will attend. In case of unforeseen circumstances, Prof Kristina Blennow (SLU) will deputise in her place. The MB will ensure the smooth and timely operation of the project and manage the associated risks. They will also be responsible for data management, including IPR management, and ensuring regulatory and ethical compliance of all aspects of the work as described in the consortium agreement. They will also have responsibility for conflict management and the resolution of minor or more serious disputes that arise in the consortium. A management plan for this will be included in the CA.

DisCover Crop will be supported by a Scientific and industrial advisory board that includes members from academia and industry with expertise in relation to multi-actor projects and the use of cover crops in European farming systems (Fig 6).

3.2 DisCoverCrop Workflow

The DisCoverCrop workflow is shown in Fig6a. Using the workplan and CA documents as a starting point, we will develop and agree on joint protocols for experiments and co-design field experiments, taking input from project partners, stakeholders, advisors and including external advice. After data collection and analysis (carried out in WP2 and WP3, as presented in Fig 3), we will co-evaluate the outputs and adjust experimental protocols before repeating the process for the next field season (central circle Fig 6a). In this way we will utilize our field trials as living labs. In a parallel process, we will carry out communication-needs research with both experts and stakeholders such as farmers and advisors (WP4, Fig 3) to identify farmers' needs, decision-making strategies and local knowledge about the use of cover crops and sustainable potato production. We will also co-evaluate the outputs of WP4 and connect these to WP2, WP3 and WP5 before redesigning the next field trials. The outputs from the research carried out in WP2-4 will be synthesized to develop new practical guidelines for the use of cover crops in agroecosystems (WP5, Fig 3). An overview of the connections between the WPs is presented in Fig 6b.

Each scientific WP is designed to function independently of, but in connection with, the other WPs and are all carried out transnationally. In this way we mitigate the risk that failures in one WP disrupt another WP, whilst maximizing the benefit of our multi-actor interdisciplinary transnational approach.

3.3 Implementation of field trials and Living Labs

In DisCoverCrop we will use field trials in a Living Labs approach, combined with Virtual Living Labs to connect practitioners transnationally. The use of Living Labs is a method to bring multiple stakeholders together in a learning environment to test ideas and concepts such as new technologies or practices (Potters et al. 2022, Gamache et al. 2020). The aim is to test the concept in a 'real-life' situation in a co-creative rather than top-down expert led manner (Gamache et al. 2020). The inclusion of the farmers in the co-created guidelines means that the outcome should be actionable in a range of situations by other farmers, as the results are grounded in their working experience and embraces complexity (Potters et al. 2022). This approach therefore supports a sustainable transition, built on flexibility and experimentation, that is also at the same time more responsible and respectful of the actors involved (Gamache et al. 2020, Potters 2020).

Field trials, both managed and led by research partners (research trials; RT) and those managed and led by farmers (farmer trials; FT) will be carried out in each partner country (see tables at the start of the workplan). In SE, NL, NO and DE at least one RT or FT site has already been established and either preliminary or longer-term data from cover crop trials is already available as background to the project. In SE Lyckeby Starch AB, have started testing the effects of various cover crops on potato performance and yield in an FT site. This data is available to the project participants and will be built on in DisCoverCrop to provide more detailed scientific data on the performance of both the cover crops and potato. In NL, NO and DE the established sites are research sites with a long history of expertise in the management of field trials. The RT and FT in Germany are conducted and managed by the Dethlingen Research Station, which is under the control of the Lower Saxony Chamber of Agriculture. The Lower Saxony Chamber of Agriculture plays a crucial role in supporting agricultural development across the country by overseeing the research stations, including Dethlingen and acting as an agricultural advisory service. It provides resources and expertise to improve agricultural practices, with a particular focus on potato research and development. An advisory board representing the funding partners steers the work program, while the Lower Saxony Chamber of Agriculture is responsible for contractual implementation, making the Dethlingen research station an important resource for farmers and agronomists in the region.

RT in Norway are conducted by NORSØK which owns the certified organic farm "Tingvoll Gard" in Tingvoll. In 2023 a long-term field trial with cover crops in a narrow potato cropping rotation was established and will be continued throughout the project phase, all data will be available. The RT is designed as a living laboratory where the researchers work closely with farmers to demonstrate agroecological practices. Results will be disseminated together with advisors from NLR (Norsk Landbruksrådgiving) which guides farmers across Norway. The Norwegian FT will be performed together with farmers which already stated their participation in the project in the form of letters of intent. Field trial plans will be co-created with scientists, farmers and advisors at all stages of the project.

WUR has a number of field trial sites across NL, which work according to PD-certified guidelines and are GLOBAL-GAP certified. These sites are managed RT sites allowing practice-orientated research with equipment and expert personnel on site to support R&D work in a co-creation approach.

Meetings including members from across the consortium, will take place before, during and after the field seasons (both for the cover crops and potato) to develop shared protocols, analysis tools and to share findings to allow co-creation and co-evaluation of research in an iterative multi-actor approach.

CO-CREATION AND MULTI-ACTOR APPROACH

DisCoverCrop is a collaborative project bringing together potato growers, agricultural authorities and academics from different disciplines with the overall aim of promoting sustainable potato production using cover crops. Our approach is one of co-creation, with farmers (our end-users) at the heart of the project to understand the real-world impacts of cover crops on soil and potato health, and farmers' perspectives on potato production.

Our field trials in DE, NL, SE, EE and NO serve as farmer-managed living laboratories, covering several sites at a regional level and evaluated by transdisciplinary research activities. Researchers will study the performance of cover crops, analyse the soil health and pathogen suppression effects of cover crops, and relate the results to farmers' experiences and local agronomic considerations. We will establish our field trials as open innovation ecosystems in real-world environments, using iterative farmer feedback processes to ensure we are testing and evaluating the practical and economic viability of cover crops under real-world farming conditions.

Together with agricultural consultants, we will analyse the results from these real-life situations and derive agro-ecologically oriented potato cultivation practices for the climatic regions of Northern and Western Europe. By effectively using communication research tools from the humanities and the communication networks of the involved advisors (and other identified stakeholders shown in Fig 5), we will accelerate and broaden the acceptance and adoption of our results by farmers. With the active involvement of potato farmers from 5 countries at every stage, we will innovate the use of cover crops in potato production to meet real needs for more sustainable potato production in the

EU. This direct involvement ensures that we take into account the local knowledge and practical concerns of farmers and increases the likelihood that our results will be widely adopted.

DisCoverCrop Workplan - Annex V

Staff effort

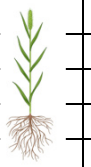
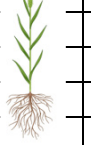
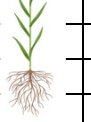
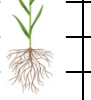
Staff effort per partner in person months for the entire duration of the project.

Partner, Institution, Country	WP1	WP2	WP3	WP4	WP5	Total
 P1 – SLU (SE)	14	17	5	10	6	52
 P2 – WUR (NL)	1	6	1	1	1	10
 P3 – NORSØK (NO)	4	2	2	3	10	21
 P4 – FZJ (DE)	1	2	29	2	2	36
 P5 – EMU (EE)	5	5	1	6	4	21
Total	25	32	38	22	23	140

Co-creational research approach

Field trials	Conventional management		Organic management	
	RT	FT	RT	FT
P1 – SLU (SE)		1*	1	
P2 – WUR (NL)	2*			1
P3 – NORSØK (NO)		1	1*	1
P4 – FZJ (DE)	2*			1
P5 – EMU (EE)	1			1

RT = Research field trial, FT = Farmer-managed trial, * trial site already established

Cover Crops used	Plant Family			
	BR	PO	FB	PT
P1 – SLU (SE)	2 		1 	1 
P2 – WUR (NL)	2 	1 	1 	
P3 – NORSØK (NO)	1 	1 	1 	1 
P4 – FZJ (DE)	1 	1 	2 	1 
P5 – EMU (EE)	1 	1 	1 	1 

BR = Brassicaceae (e.g. oilseed radish), **PO** = Grass and grains (e.g. Japanese oats, winter rye,...),

FB = Fabaceae (e.g. Trifolium incarnatum, summer/winter vetch, winter peas), **PT** = Plantaginaceae

Involvement of specific external experts/stakeholders (see further descriptions in letters of interest)	
P1 – SLU (SE)	Lyckeby Starch AB (farmer-owned cooperative and starch potato company) ^{1,3}
P2 – WUR (NL)	HZPC (Dutch seed potato trading & breeding) ^{1,2} NAO (<i>Dutch Potato Organisation</i>) ³ BO Akkerbouw ³
P3 – NORSØK (NO)	Sunndalspotet AS (potato producer & trading) ^{1,3} NLR <i>midt</i> ² Norwegian Potato Fagforum ^{2,3}
P4 – FZJ (DE)	Research Station Dethlingen (subcontractor for potato research) ^{2,3} , LWK Lower Saxony ²
P5 – EMU (EE)	Eesti Kartul ^{2,3} Talukartul ^{1,3}

¹Potato production and/or processing company, ²Agricultural Advisory, ³Trade organisation and/or farmer organisation.

Work package 1 – LEAD & IMPACT

WP1	Coordination, Communication, Dissemination				
WP lead	P1 - SLU	Contributors	P1, P2, P3, P4, P5 + local farmers	Start – End	M1 – M36
Objectives					
1) Lead and coordinate the project including exploitation, communication and dissemination, ensuring transparent and participatory communication flow across work packages and among partners, and stakeholders. 2) Act as the central contact for the project consortium for the lifetime of the project, including managing communication with the call office, and participating in joint network activities. 3) Monitor project progress to ensure all work is carried out to a high standard and that the project meets contractually bound deliverables and milestones as presented in the proposal. 4) Share all information with project partners in a timely manner and ensure punctual delivery of reports. 5) Coordinate outward-facing communication and dissemination activities so that high quality scientific evidence produced can have clear and measurable impacts for all stakeholders and beyond the life of the project.					
Description					
Task 1.1: Coordination of the consortium agreement and appointment of management boards, lead: SLU, contributors: WUR, NORSØK, FZJ, EMU. A consortium agreement (CA) will be drawn up outlining the rights and contractual obligations of each partner to the project and to the funding bodies. This work will be led by the lawyers at SLU with support from all partners. Among other things it will outline project management and decision making, how to settle internal disputes, IPR rights, and ensure regulatory and ethical compliance. In connection to this the internal management board and an external scientific and industrial advisory board will be formed and the data management plan extended/updated. Task 1.2: Project monitoring and organisation of project meetings, lead: SLU, contributors: WUR, NORSØK, FZJ, EMU. Regular meetings will be organised for the project members. These include a physical kick-off meeting at SLU, SE and a final meeting to take place at WUR, NL where research field trials can be shown to all partners. In addition, project meetings will take place every 6 months (including relevant stakeholders e.g., advisors and farmers) online where the progression of the different work packages will be discussed. To monitor project progress, the coordinator and management board (WP leaders) will meet online every 3 months, to give updates on progress, milestones and deliverables. Possible delays will be discussed as soon as they are identified and mitigation actions taken as soon as possible, following the risk and mitigation plan for each WP. Task 1.3: Communication with the call office, and participation in joint AGROECOLOGY meetings lead: SLU, contributors: WUR, NORSØK, FZJ, EMU. The coordinator will attend joint AGROECOLOGY meetings and ensure deliverables and reports are delivered to the call office in a timely fashion. If delays and bottlenecks are identified that are not solvable within the project, these will immediately be communicated to the call office to identify other solutions. Task 1.4 Communication channels and media platform, lead: NORSØK, contributors: SLU, WUR, FZJ, EMU Setup and maintenance of online platforms: I) a project website (M3) in collaboration with all partners and other stakeholders which serve for data sharing (intranet login) and the dissemination of findings generated and activities during the project phase (including news page and newsletter registration). II) a DisCoverCrop YouTube channel will be launched (M6) accessible to all interested stakeholders which will feature project activities and findings. III) social media presence will be established. Task 1.5: Coordination of project dissemination activities including webinars & workshops as exchange forums, lead: NORSØK, contributors: SLU, WUR, FZJ, EMU: Dissemination activities will be coordinated in a project dissemination database. A series of webinars will be organised targeting farmers, advisors, and policymakers, as well as networks we have identified, such as “CoverCropsNordic” and “CropDrive”. These webinars will aim to foster and accelerate information exchange between multi-network groups, promoting the best cover crops for different regions and various cropping scenarios. Our dissemination activities (according to the DEC) will include field days and practical workshops where farmers are invited for exchange and discussion. Through virtual living labs and virtual farms (in collaboration with WP4, D4.6), we will provide access for farmers across Europe to participate in online workshops on best cover crop practices. This platform will facilitate bilateral knowledge exchange between					

Workplan – DisCoverCrop

scientists, European farmers and advisors. This task also includes coordinating production of project factsheets, synthesizing project outcomes in local languages.	
Deliverables	D1.1 – Consortium agreement drawn up and signed by all partners (M1) – SLU D1.2 – Database of project stakeholders established (M4) – SLU D1.3 – Extended, updated Data Management Plan (M5) – SLU D1.4 – Websites and social media channels launched (M6) – NORSØK D1.5 – Short films of DisCoverCrop objectives and results produced (M30) – NORSØK D1.6 – Factsheets in English & local languages synthesizing project results (M34)– SLU D1.7 – Policy brief synthesizing project outputs and outcomes (M35) – SLU D1.8 – MTR (M18), final report (ETR) (M36) & interim reports to national funders – SLU
Milestones	M1.1 – Formation of management & scientific and industrial advisory boards (M3) – SLU M1.2 – Kick-off meetings (Digital pre-kick-off M1 & physical kick-off M4) Alnarp, SE-SLU M1.3 –Webinars and workshops organised (M6, M18, M30) – NORSØK M1.4 – Coordinator attendance at three joint AGROECOLOGY network meetings – SLU M1.5 – Final meeting held in WUR, NL – WUR (M35)
Risk and mitigation of risks	Risk 1.1 – Coordinator unable to continue in project for some reason. Mitigation: Another team member from coordinating institution takes over coordination activities. Risk 1.2 – Dispute or disagreement arises between partners. Mitigation: Coordinator LGB has training in conflict resolution and the resolution plan laid out and agreed upon in the CA will be followed to resolve issues as soon as they arise. Risk 1.3 – Online dissemination activities are not visible to farmers. Mitigation: To increase visibility of the online dissemination activities of the DisCoverCrop project we will actively advertise our activities through known and established communication channels of farmers in each country. Via our online platforms (website and YouTube channel) stakeholders have continuous access to the project results. Risk 1.4 – Dissemination of findings is not accessible for farmers not speaking English. Mitigation: All dissemination material and activities will be available in the languages of the participating countries of the project members and in English. Risk 1.5 – Delayed analysis might delay the deliverable reports with deliverable dates close to the project end date and this might affect the final synthesis project report in M36. Mitigation: Coordinator will follow up on the deliverable progress of each partner closely. Any serious delay will be reported to the project board and the funding bodies.

Work package 2 – BIOLOGICAL ASSESSMENTS

WP2	Effects of cover crops on biotic interactions				
WP lead	P2 - WUR	Contributors	P1, P2, P3, P4, P5	Start – End	M7 – M34
Objectives					
1) To assess the effect of cover crops on soil bacterial and fungal biomass and community composition and on soil nematode community composition. 2) To determine the effect of cover crops on disease suppressiveness and pathogen abundance in a potato cash crop. 3) Uncover one potential mechanistic effect of cover crops on disease suppression by association of soil microbial and nematode community parameters with suppressiveness and pathogen abundance.					
Description					
Cover crops are known to have several positive effects on the following crop. There are indications that cover crops or the previous crop can contribute to the delay or reduction of important above- and below-ground diseases, such as early blight (<i>Alternaria solani</i>), late blight (<i>Phytophthora infestans</i>) and black scurf (<i>Rhizoctonia solani</i>), Verticillium wilt or root knot and root lesion nematodes respectively (Griffin et al. 2009, Runno-Paurson et al. 2019). Currently, some of those diseases are controlled by the use of pesticides, some of which will be restricted in use in the coming years. For others, such as problems caused by nematodes, farmers (in NL) are already experimenting with crop rotations and the use of certain cover crops that are directly detrimental to some nematode species. A number of studies have indicated that changes in the soil microbial community, biomass or activity can elicit soil suppressiveness against one or more pathogens and pests (Schlatter et al. 2017, Kurm et al. 2023). It has long been known that the type					

of crop (main crops and cover crops) has a substantial influence on the soil microbial community, mostly through root exudation of nutrients and secondary metabolites. The biomass of cover crops incorporated into the soil increases populations of indigenous bacteria and fungi suppressing several potato pathogens (Abbasi et al., 2019). However, also specific microorganisms are known to have antagonistic effects against certain pathogens and can be used as biocontrol agents contributing to a reduction in pesticide use (Gorai et al. 2021, Hicks et al. 2014). Still, these biocontrol agents are usually used individually and have not been studied in combination with other treatments to reduce disease such as cover crops. Survival of inoculated biocontrol agents in open fields is poorly understood, but we hypothesize that green bridges such as those formed by the use of cover crops may facilitate survival of biocontrol agents in potato production systems. In addition, both the microbial and the soil nematode community (as opposed to only plant parasitic nematodes) have been successfully used as an indicator for soil health. These communities have repeatedly been found to be related to decomposition and nutrient cycling and have been reported to be affected by cover crop identity (DuPont et al. 2009). However, their role in potato cultivation is still unclear. WP2 focuses on the effect of cover crops on soil microbial and nematode communities. These parameters will be related to soil suppressiveness and above- and below-ground plant diseases. In addition, the effect of the addition of specific biocontrol agents against potato diseases in combination with cover crops will be assessed. Moreover, the association of microbial and nematode community parameters with soil abiotic parameters (studied in WP3) will be analysed. This WP will forge a link between cover crop use, soil microbial life, soil nutrient cycling and soil-borne diseases.

Task 2.1: Soil microbial communities, lead: WUR, contributors: SLU, NORSØK FZJ, EMU. There is extensive evidence that microbial community composition, biomass and activity is related to suppressiveness against soil pathogens. However, the influence of the identity and management of cover crops before potato production on the microbial community and consequentially suppressiveness is poorly understood. Therefore, in task 2.1 soil samples from the field experiments will be used to analyse both the bacterial and fungal biomass by Phospholipid-derived fatty acids (PLFA) analysis and/or metabarcoding.

Task 2.2: Soil nematode communities, lead: WUR, contributors: SLU, NORSØK, FZJ, EMU. The soil nematode community composition is correlated with nutrient cycling and decomposition and therefore an important indicator of soil health. In task 2.2 the community composition of environmental nematodes will also be assessed and related to other measured parameters, such as soil nutrient status (WP3) and crop performance (WP5).

Task 2.3: Evaluation of biocontrol agents, lead: EMU, contributors: SLU. Efficacy of biocontrol agents in suppression of potato foliar and tuber diseases will be assessed in small-scale field trials in EE and SE in this task. Application of granular preparations based on *Trichoderma harzianum* and *T. asprellum* in planting of potatoes will be tested for their action against tuber infections caused *Rhizoctonia solani* and *Helminthosporium solani*. Foliar application of products based on *Pythium oligandrum*, *Bacillus subtilis* and chitosan will be screened for reduction of late and early blight infection. Preparations will be applied to cover crops their effects on cover crop performance assessed. The ability of individual cover crops to host biocontrol agents overwinter in the rhizosphere will also be assessed. Finally, the addition of Biocontrol agents in potato after use of cover crops will be tested to study their synergistic effect in disease control.

Task 2.4: Disease suppressiveness, lead: WUR, contributors: SLU, NORSØK, FZJ, EMU. The effect of cover crops on disease pressure and suppressiveness will be assessed for several relevant potato pathogens. Soil samples (fresh, stored at 4°C) from the field experiments will be used to assess disease suppressiveness against *Rhizoctonia solani* and *Pythium ultimum* in a bioassay setup established at WUR. Since this assay can only be performed on a selected number of samples due to logistic limitations, in the remaining fresh soil samples the abundance of *Rhizoctonia solani* will be determined using a quantitative molecular assay.

Task 2.5: Quantify soil and leaf pathogens, lead: SLU, contributors: WUR, NORSØK, FZJ, EMU, local farmers. For several pathogens a link has been established between abundance in soil and the likelihood or amount of disease caused in the crop. For instance, damage thresholds have been identified for plant parasitic nematodes in several crops. For countries in which plant parasitic nematodes are relevant for potato farming soil samples will be subjected to a quantitative analysis of plant parasitic nematodes by microscopic counting. Leaf pathogens on potato will be quantified through observations in the field. The results of these assays will be related to cover crop use and management.

Task 2.6: Data analysis, lead: WUR, contributors: SLU, NORSØK, FZJ, EMU. Eventually, cover crop treatment, microbial parameters and disease suppressiveness will be analysed together to find associations indicating a mechanistic effects of cover crops on suppressiveness. In addition, soil microbial and nematode community composition will be analysed together with abiotic parameters measured in WP3.

Deliverables	D2.1 – Analysis and report of the effects of cover crops in the different field trials on soil biological parameters – WUR D2.2 – Report on the development of above- and belowground diseases after the cultivation of different cover crops in the participating countries – WUR D2.3 – Report on the effect of biocontrol agents early and late blight- EMU D2.4 – Statistical analysis of the association of soil suppressiveness, soil biotic and abiotic parameters and above- and belowground disease – WUR
Milestones	M2.1 – Finished protocol for field sampling – WUR M2.2 – Finished protocols for biological parameter analysis – WUR M2.3 – Completion of first field sampling & disease suppressiveness bioassay – WUR M2.4 – Completion of second field sampling & disease suppressiveness bioassay – WUR M2.5 – Finished measurements of biological parameters from first sampling – WUR M2.6 – Finished measurements of biological parameters from second sampling – WUR M2.7 – Lab and controlled environment testing of biocontrol agent activities on cover crops and against potato diseases – EMU
Risk and mitigation of risks	Risk 2.1 – Natural disease pressure can be high, which can obscure effects on suppressiveness. Ideally, sites will be chosen that are not known for extremely high pathogen abundances. Risk 2.2 – Microbial communities often show high variation between countries/locations, which makes it challenging to discover effects of treatments by statistical methods. Therefore, care will be taken to establish full-factorial experiments with sufficient numbers of replicates in each country in the research field trials.

Work package 3 – PHYSIO-CHEMICAL ASSESSMENTS

WP3	Effects of cover crops on abiotic parameters				
WP lead	P4 - FZJ	Contributors	P1, P2, P3, P4, P5	Start – End	M2 – M33
Objectives					
1) To evaluate the impact of cover crops on soil physico-chemical properties and soil aggregate structure. This objective focuses on understanding how cover crops influence key soil properties such as bulk density, pH, elemental composition, or organic C content. By examining changes in the soil aggregate structure and stability, we aim to identify how cover crops control erosion, improve water infiltration, root penetration, and nutrient retention. 2) To investigate the effects of cover crops on nutrient cycling and availability in top- and subsoil layers and nutritional status in leaves. This objective aims to assess the role of cover crops in mobilizing nutrients from deeper soil layers and enhancing nutrient use efficiency. By studying movement and availability of essential nutrients like N, P, and K we seek to understand how cover crops contribute to improved soil fertility, reduced nutrient losses, resource allocation in plants, and increased yields.					
Description					
The physico-chemical properties of soil, such as texture, density, pH, elemental composition, and organic matter content, are fundamental determinants of soil health and productivity. These properties influence soil microbial communities (linked to WP2 Task 2.1: Soil microbial communities), which are essential for nutrient cycling and organic matter decomposition. Cover crops, particularly those with deep rooting systems, can modify these physico-chemical properties, enhancing soil fertility and structure. These modifications can lead to increased microbial biomass and activity, improving nutrient availability and soil health (Bauke et al., 2017; Barber, 1995). Deep-rooting plants can penetrate and modify soil structure at greater depths, creating biopores and channels that improve soil aeration, water infiltration, and root growth of subsequent crops. However, the ability of roots to penetrate deeper soil layers is often limited by increasing soil strength with depth, which confines deeper roots to existing pore networks. In addition,					

cover crops have a great impact on the soil's physical structure, specifically soil aggregates. Soil aggregate structure is a crucial indicator of soil health, affecting water infiltration, root penetration, and nutrient retention. The stability of soil aggregates is enhanced by organic matter, which is often increased by the presence of cover crops. The impacts on deep-rooting cover crops on soil directly influence soil aggregate stability (Koch et al., 2021). Improved aggregate stability has been linked to enhanced soil health and resilience against erosion and compaction (Koch et al., 2021). Deep-rooting plants contribute to soil health by increasing soil organic matter at depth through root exudates and dead root tissue. Building on the understanding of soil structure, we will investigate how these structural changes affect nutrient dynamics within the soil. Nutrient cycling within the soil is essential for maintaining soil fertility and supporting sustainable crop production. Cover crops, especially those with deep root systems, can play a significant role in enhancing nutrient cycling by mobilizing nutrients from deeper soil layers and making them available to subsequent crops (Chen and Weil, 2011; Bauke et al., 2017). These crops can improve the retention of essential nutrients like N, P, and K, reducing leaching losses and enhancing nutrient use efficiency (Koch et al., 2021). When deep roots decompose, they release these nutrients back into the soil profile, potentially making them available for future crops. This process can help recover nutrients that have leached below the typical rooting zone of annual crops, reducing nutrient losses and improving overall nutrient use efficiency. The presence of biopores created by cover crops can also enhance microbial activity (linked to WP2 Task 2.1: Soil microbial communities) and nutrient turnover, leading to improved soil fertility (Hoang et al., 2016).

Task 3.1: Physico-chemical soil properties, lead: FZJ, contributors: SLU, WUR, NORSØK, EMU

Topsoil as well as subsoil (under plough layer) will be sampled in already established field experiments in DE, SE, and NO with cover crops and/or subcontracted farmers that plant cover crops with the guidance of the partner in the respective country. Sampling timepoints are in the spring before potato planting and at potato harvest. Soil physio-chemical properties from all samples will be measured at FZJ. By analysing these samples, we aim to understand the extent to which cover crops can alter soil physico-chemical properties and the subsequent effects on soil health and productivity.

Task 3.2: Soil aggregate structure, lead: FZJ, contributors: SLU, WUR, NORSØK, EMU

This task will focus on assessing the impact of cover crops on soil aggregate stability and structure. We will determine the aggregate size distribution as well as aggregate stability of water-stable aggregates using laser diffraction (Siebers et al., 2023).

Task 3.3: Soil nutrient cycling, lead: FZJ, contributors: SLU, WUR, NORSØK, EMU

In this task we aim to investigate the effects of cover crops on nutrient cycling, focusing on how these plants influence the movement and availability of nutrients in both topsoil and subsoil layers. We will analyse total and plant available nutrient contents in the top- and subsoil (e.g., N, P, K) of all field trials in the respective countries for selected time-points.

Task 3.4: Potato leaf nutrient assessment, lead: NORSØK, contributors: WUR, SLU, FZJ, EMU

The content of macronutrients (N, P, K, Mg, S, and Ca) of potato plants in the field trials will be assessed mid-season in all partner countries by the respective partners. Samples will be sent to a subcontracting laboratory (EUROFINS). Results will serve to check potato health, resource allocation and stress/disease susceptibility, since foliar diseases such as early blight are known to be correlated with nutrient deficiencies and farm management strategies (Stridh et al 2024).

Deliverables	D3.1 – A report summarizing the physico-chemical properties of soil samples collected from field experiments for different time-points in DE, NO, NL, SE, and EE. This report will include data on soil texture, density, pH, elemental composition, and organic matter content. (M3.3) – FZJ D3.2 – A report of a comprehensive assessment of soil aggregate structure including the size distribution and stability of water-stable aggregates for all field trials in in DE, NO, NL, SE, and EE. The report will detail the impact of cover crops on soil aggregate stability and structure, highlighting improvements in soil structure, aeration, and resistance to erosion. (M3.3) – FZJ D3.3 – A report describing the effects of cover crops on nutrient cycling within the soil. This will include analysis of total and plant-available nutrient contents (N, P, K) in both topsoil and subsoil layers, focusing on how cover crops influence the movement and availability of nutrients. (M3.3) – FZJ
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	D3.4 – A report on potato plant leaf macronutrient status in the field trials including relations to growth properties and the soil nutritional status (stated in D3.3) as well as the disease intensity (stated in D2.2) – NORSØK
Milestones	M3.1 – Protocol for soil sampling for physico-chemical analyses – FZJ M3.2 – Preparation of the soil samples taken for aggregate structure and stability measurements for all partners – FZJ M3.3 – Completion of nutrient content analysis in soil samples from all partners – FZJ M3.4 – Protocol for sampling, completion of sampling and shipping of potato leaf samples for nutrient analysis to an external commercial laboratory – NORSØK
Risk and mitigation of risks	Risk 3.1 – Potato tuber harvest might destroy newly formed aggregates, which might disguise beneficial effects of cover crops on soil aggregate structure. Mitigation of this risk is a sampling before and after cover crop growth and before and after potato growth and harvest.

Work package 4 – FARMER COMMUNICATION RESEARCH

WP4	Cropping behaviour and co-creating evidence-based communication guidelines				
WP lead	P5 - EMU	Contributors	P1, P2, P3, P4, P5 + local farmers	Start – End	M1 – M34
Objectives					
The objectives of this work package are to 1) understanding stakeholder cropping behaviour and 2) to co-create evidence-based communication guidelines to support both co-creation of research within DisCoverCrop, and the implementation of the resulting research results in real-life situations within and beyond the lifetime of the project.					
Description					
The dynamics between scientific and practice-based evidence vary across cultures and professions. Globally, there is growing interest in integrating practice-based and scientific knowledge, with increasing attempts for how integration can be achieved. For example, in SE, education, institutions, and professions have long been guided by the concept of science and proven experience (Persson et al. 2019; Persson 2021). Local "proven experience" can shape worldviews as much as scientific evidence, influencing behaviour beyond regulations (Persson et al. 2019; 2022). Comparing local "proven experience" with expert knowledge reveals a crucial gap between what decision-making agents know and what they need to know to improve decision-making and fulfil their goals (Ingram 2008; Fischhoff et al. 2011; Palmér et al. 2023). The Swedish notion of science and proven experience is unique, offering a novel international contribution. The fact that agriculture has seen fewer efforts than medicine and health care to apply this approach indicates potential for new insights and solutions in this project. By bringing different stakeholders together in a Living Lab approach creates bridges between the different knowledge (Gamache et al. 2020), therefore in this WP, practice, science and socio-economic influences will be taken into account together. Task 4.1: Identifying critical facts farmers and experts need to know about potato cultivation, cover crops, their management and the policy setting, lead SLU, contributors: WUR, NORSØK, FZJ, EMU. This task will take part in several stages by creating influence diagrams - a well-established method for the cognitive mapping of a decision problem - conducted in various workshops (see Wood et al. 2012). The workshops will engage scientists, farmers, cultivation advisors and other interested stakeholders, to compare the influence diagrams of different groups and to identify knowledge gaps between local knowledge and scientific knowledge. The experts in the consortium will conduct one workshop at the first consortium meeting, then approx. 8 farmers and up to 5 agricultural advisors/interested stakeholders in each country will be invited to further workshops (physical/online). Task 4.2: Large-scale survey of communication needs regarding potato cultivation and cover crops, lead SLU, contributors: WUR, NORSØK, FZJ, EMU. Building on the knowledge gaps identified in T4.1, surveys will be sent to approx. 90% of the potato farmers in each country to identify their communication needs related to potato cultivation and cover crop use. Farmers will be invited via e-mail to					

participate in a web-based survey utilising various channels such as advisory boards, cooperatives, the networks across the consortium etc. The questionnaire responses will be analysed to identify communication needs based on identified drivers of their decision-making and decision-making strategies. using data collected on beliefs, values, management measures, local knowledge, expectations of different potato management systems with and without cover crops and decisions taken (*cf.* Blennow et al. 2020). The machine learning modelling method of Bayesian Additive Regression Tree modelling (Kapelner and Bleich 2016) will be used to analyse the survey data, *cf.* Blennow et al. 2020.

Task 4.3: Virtual tools to support farmers in dispersed populations, lead: EMU, contributors: SLU, WUR, NORSØK, FZJ. Virtual living labs and virtual farm tours will be explored as two methods to bring dispersed populations together with geographic challenges e.g. SE, NO and EE. A virtual farm tour will also allow the consortium to demonstrate the impacts of cover crops on potato production to those unable to travel to a demonstration farm. One virtual living lab and one virtual farm tour with subtitles translated into the different languages of the consortium will assist in disseminating information across countries as well as across distances within countries.

Task 4.4: Co-created guidelines for effective communication on potato cultivation and the use of cover crops, lead SLU, contributors: WUR, NORSØK, FZJ, EMU. Guidelines for effective communication with farmers and their advisors will be co-created with the project reference group in a series of workshops. The communication needs identified in T4.2 will be used to co-create guidelines for effective communication with farmers on the use of cover crops in the potato cultivation system. The approach adheres to a recently developed rationale and methodology that has proven successful in developing guidelines for effective communication with forest owners and professionals regarding adaptation of forestry to climate change (see, e.g., Blennow et al. 2020; Persson, Blennow et al. 2020; Blennow and Persson 2021; Blennow et al. 2021).

Deliverables	D4.1– Report on the decision-problem mapping and knowledge gaps on potato cultivation and cover crops (M4.1, 4.2 & 4.3) – SLU (M8) D4.2 – Survey Report (M4.4-M4.5) – SLU (M32) D4.3 – Virtual living lab group created in each country with dispersed rural populations – EMU (M3) D4.4 – Virtual Farm Tour – EMU (M34) D4.5 – Co-created guidelines for effective communications on potato cultivation and cover crops – SLU (M34)
Milestones	M4.1 – Scientists’ influence diagram – SLU M4.2 - Stakeholder’ influence diagrams completed – EMU M4.3 – Dispatch of survey to farmers – SLU M4.4– Analysis of survey complete - Set of identified famer and advisor knowledge-needs – SLU M4.5 – Living Labs set up – EMU M4.6 – Virtual Tour inputs (videos, podcasts, photos etc.) – EMU M4.7 – Virtual Tour webpage complete – EMU M4.8 – Co-creation workshops completed to develop guidelines effective communications on use of cover crops in potato cultivation — SLU M4.9 – Co-creation workshops analysed – SLU
Risk and mitigation of risks	Risk 4.1 – There is a risk that farmers and advisors are not comfortable with sharing information for analysis in a workshop format. In this case interviews will be conducted to map the decision problem of potato cultivation with cover crops. Risk 4.2 – Too few survey responses for the analysis. Extra stakeholders will be sent surveys, where possible, so a critical number of responses is still available. Reminders will also be issued to invitees. Alternative analysis methods will be used if few respond. Risk 4.3 – Despite the use of virtual tools to overcome issues of distance there is a risk that the Internet connectivity also provides a barrier to obtaining information, accessing the virtual living lab or being able to take a virtual tour of a farm. To overcome this low bandwidth resources will also be made available, such as pdfs connected to the interactive

web-page tour. The videos and podcasts will also be kept short to reduce the burden on the internet use.

Work package 5 – AGROECOLOGICAL MANAGEMENT

WP5	Agroecological management practices and estimations for potato management				
WP lead	P3 -NORSØK	Contributors	P1, P2, P3, P4, P5, local farmers	Start – End	M3 – M36
Objectives					
1) Primary objective of WP5 is to co-design farmer-managed field trials, monitor their implementation, and assess the performance of cover crops and the following crop together with the farmer. 2) Secondary objective is to synthesize and translate the results in recommendations (best practices) on the use of cover crops e.g., in potato production for each partner country, which will then be disseminated (linked to WP1).					
Description					
The prevailing approach to potato production, driven by economic considerations, has resulted in practices that have led to adverse effects on soil properties, such as soil compaction, erosion, reduced water infiltration and soil organic carbon losses. These factors have the potential to negatively impact potato productivity (Stark & Porter 2005; Larkin et al. 2021). The implementation of agroecological approaches to potato production, such as the incorporation of cover crops has the potential to mitigate risks, enhance the physical, chemical, and biological properties of the soil (Magdoff 2001), and sustain or enhance yields (Rittl et al., 2023). In WP5, the extent of the benefits of cover crops (in agroecological terms – increased plant diversity) across different climatic regions and soil types will be tested, evaluated (in organic and conventional systems), and a comparison will be drawn between these benefits and the potential risks to the potato grower of adopting these cover crop strategies. For example, the selected cover crops may have a negative impact on the performance of the potato crop (allelochemical plant-plant interactions). If negative effects are evident, consideration will be given to reducing the impact of the cover crop on the crop, e.g., by earlier ploughing or by excluding the species. The adoption of cover crops in crop rotations varies considerably between farmers and countries, as the profitability of such strategies is contingent upon management decisions that directly impact productivity at the farm level (Snapp et al. 2005). The motivation behind potato farmers' initiatives is primarily economic, often missing the focus on the long-term sustainability of the farm and the reduction of input requirements. Consequently, there is a pressing need for knowledge on cover crop management, a proof for their usability, as well as an understanding the agroecological benefits for long-term profitability and sustainability, as well as of the economic costs at the farm level. This knowledge is an essential asset for new agroecological farming strategies that are easily understood and accepted by European farmers. This WP will link the biological (WP2) and physico-chemical (WP3) soil effects of cover crops with the practical and economic considerations and concerns at farm-level. Task 5.1: Co-design strategies for sustainable potato production, Lead: NORSØK, contributors: SLU, WUR, FZJ, EMU Taking advantage of the ongoing experimental research trials in NO, DE and NL, the most suitable cover crops for potato production in the regions will be analysed and the farmer-managed trials will be co-designed with relevant stakeholders in both organic and conventional farming systems, including both early and late potato varieties. This will involve discussing the type of cover crop and its management within potato production and developing a best practice design plan for each individual farm ("case study") and cover crop to be implemented in the field trial protocol. This will be based on a whole farm health principle: best practice for the health of soil, nature and the farming community, taking into account how cover crops fit into existing farm activities and the adaptive capacity of the farming community. FTs according to the plan will be carried out within all partner countries. Task 5.2: Performance of cover crops, Lead: SLU, contributors: NORSØK, WUR, FZJ, EMU All partners will evaluate the performance of cover crops sown before the potato crop according to a common protocol in terms of establishment, survival and interactions between species in cover crop mixtures. Different cover crop performances per country will be compared and conclusions will be drawn. Task 5.3: Economic and potato quality assessment, Lead: NORSØK, contributors: SLU, WUR, FZJ, EMU For all trials (RT and FT), tuber yield and quality measurements will be carried out. A joint protocol for the evaluation will be drawn up and all data collected will be reported. The total weight of fresh tubers is recorded in the field and corrected for the number of plants per plot. Quality tests (e.g., tuber dry matter)					

and a potato health screening will be carried out each year at harvest and 6 months after storage. The assessments will include tuber size distribution and the presence of various diseases or defects. The percentage of potatoes in storage damaged by diseases such as early blight (*Alternaria solani*), late blight (*Phytophthora infestans*) and black scurf (*Rhizoctonia solani*), verticillium wilt or root knot and root lesion nematodes, but also with microcracks, green skin, blemishes, hollow heart, mechanical damage and deformities will be determined. The data from the field trials will be summarised in a report and the economic benefits of using cover crops for potato quality will be assessed.

Task 5.4: Syntheses and recommendations for the use of cover crops, Lead: NORSØK, contributors: SLU, WUR, FZJ, EMU Based on the results from WPs 2, 3 and 4 with feedback from stakeholders (farmers and advisors), this task will develop a guide to best practice for the use of cover crops in potato production. This will consider economic considerations such as the overall yield but also the effect of cover crops on above and below ground disease and soil health, and the ability of the farm to adapt to the new developed strategy. Farm studies will be presented as "case studies" per country and climatic zone, recommendations will be made on how and when to use cover crops to improve one or more of these points, and the economic and time implications will be highlighted.

Deliverables	D5.1 – Summary report of two year of co-creational farm field trials – NORSØK (M30) D5.2 – Final report on the cover crop performance in the field trials over 2 years from all partners (including RT and FT) – NORSØK (M31) D5.3 – Summarizing report of potato yield data, storage qualities and disease screening across all partners. – NORSØK (M33) D5.4 – Best cover crop practice guidelines summarised in a report and translated in all partners languages – NORSØK (M34)
Milestones	M5.1 - Establishment of farmer managed field trials (FT) (M3) – NORSØK M5.2 – Data gathering of annual cover crop performance analyses (M14, M27) - NORSØK M5.3 – Summarizing data set of economic benefits and potato quality analyses across all partners (M32) - NORSØK M5.4 – Annual data collection for the preparation of recommendations based on synthesised knowledge and results (M18, M30) - NORSØK
Risk and mitigation of risks	Risk 5.1 – Delayed establishment of the cover crop due to detrimental weather conditions and labour constraints by the farmer. Mitigation: Long-term planning including the weather situations in advance. This will also allow the farmer to be prepared and act with short notice. Risk 5.2 – Cover crop establishment due to early winter or other detrimental conditions in autumn will be insufficient and low biomass with no soil coverage is present during winter. Mitigation: Cover crop species will be chosen according to the climate conditions present in each country. Germination tests before the seeding will provide confidence of the germination rate in the field.

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DisCoverCrop Dissemination, Exploitation and Communication (DEC) Plan

The dissemination, exploitation and communication of results are key objectives for the DisCoverCrop project. A summary of the DEC plan is presented in the table below, in which we strive for as much stakeholder involvement and equitable co-creation of knowledge as possible.

DisCoverCrop Output	Target Groups	Aim of dissemination	Dissemination route
Statistical analysis & reports on soil & above ground effects of cover crops on biological parameters	Farmers, advisors, breeders, scientific community	To provide state of the art information on current usage of cover crops	DisCoverCrop website, farmer magazines, scientific articles, conferences, social media, Horizon results platform
Statistical analysis & reports on the effects of cover crops on soil physico-chemical properties, structure & plant/soil nutrient status	Farmers, advisors, breeders, scientific community	To provide state of the art information on current usage of cover crops	DisCoverCrop website, farmer magazines, peer reviewed scientific articles, conferences, social media, Horizon results platform
A map of decision problems of potato farmers with influence diagrams	Farmers, advisors, scientific community, policy makers	To bring farmers views to academics and scientists and vice versa. To facilitate co-creation of knowledge and guidelines for the future use of cover crops	Stakeholder workshops, virtual meetings, DisCoverCrop website, social media, farmer magazines, conferences, scientific articles, Horizon results platform
Virtual living labs established	Farmers, advisors, scientific community	To co-design experiments co-create new knowledge on use of cover crops. To share knowledge & experience, to demonstrate use.	Open field days, virtual meetings & demonstrations, social media, DisCoverCrop website, farmer magazines, Horizon results platform
A synthesis of the new knowledge of the agroecological benefits of cover crops in potato production systems	Farmers, advisors, breeders, scientific community, policy makers, certifying bodies, value chain actors, the public	Awareness raising, presentation of the benefits of using cover crops in potato production systems	Stakeholder workshops, virtual meetings, DisCoverCrop website, social media, farmer magazines, conferences, scientific articles, factsheets, Horizon results platform, short film
Practical guidelines for the use and effective communication on cover crops for potato farmers in different locations to increase awareness if agroecological practices	Farmers, advisors, breeders, scientific community, policy makers, regulatory bodies, value chain actors, the public	Provide guidelines for practical use and effective communication of cover crops in potato production, and support policies for their use	Policy brief, (virtual) stakeholder meetings, DisCoverCrop website, social media, conferences, farmers magazines, Horizon results platform/ booster, EU Innovation radar

Explicit plans for stakeholder involvement

To maximise impact, we have mapped relevant stakeholders using a quintuple helix approach (Fig. 5). Our strategy involves participatory engagement with stakeholders, including farmers and extension workers, to co-design field trials, co-create results, and develop new management practices and practical guidelines for cover crops. We have already secured significant stakeholder support, as shown by letters of interest, and will establish a stakeholder database at the Projects start. Our guidelines will target end users, including those who contributed to their development. Stakeholders will play a key role in co-creating knowledge in WP4- 5. National stakeholder workshops will be held in local languages to co-create influence diagrams to understand local knowledge, stakeholder values and expectations. Corresponding influence diagrams for scientific knowledge will be constructed by the project scientists. By comparing farmers' and scientists' influence diagrams, knowledge gaps in farmers' scientific knowledge and gaps in scientists' local farmer knowledge will be identified. The knowledge gaps that are considered most important to fill will be jointly identified by the scientists and the project stakeholder committee. The prevalence of the identified key knowledge gaps among farmers in each participating country will be surveyed using web-based questionnaires. The surveys will also collect data on farmers' decision-making in relation to potato cropping and cover crops, which will be used to identify the drivers of decision-making and the decision-making strategies of different groups of farmers in relation to the use of cover crops in potato cropping. Farmers' group membership is an emergent property of the data. With the drivers of decision making and decision strategies identified for groups of farmers, communication needs for the different groups can be identified. These will be used in the design of guidelines for effective communication with farmers on the use of cover crops in potato production. The establishment of virtual living labs featuring overviews of the field trials in five countries will facilitate international knowledge sharing (videos, pictures and condensed research results). Factsheets and articles in industry magazines will be produced in local languages to ensure effective stakeholder access.

Exploitation plans for project outcomes and how they contribute to project impact

By the end of the project, we will have developed practical guidelines for using cover crops. Our goal is to reach as many target audiences as possible (Fig. 5) to accelerate the adoption of cover crops in potato cropping systems and create a platform for identifying exploitable data, products, and services from DisCoverCrop. To further develop DEC, we will use the Horizon results platform, produce a short film about DisCoverCrop and use the Horizon results booster, standardisation booster and innovation radar services that support Horizon Europe projects. In this way, we will connect our results directly with EU investors, innovators and policy makers to ensure that we develop strong links to maximise the exploitation potential of DisCoverCrop. These services are free of charge and therefore do not require any additional budget. All participants will dedicate at least one person-month to DEC tasks, with the coordinator allocating time for coordination of DEC (see budget for details). The coordinator will also present results at Agroecology Europe joint network meetings alongside DisCoverCrop meetings. These efforts will ensure the outcomes and impacts in Fig. 5 are achieved and that project results are fully leveraged.